

ATQASUK, AK, USA

WMO: 702685

Lat: 70.467N

Lon: 157.436W

Elev: 29

StdP: 100.97

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 27518

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.0	-38.7	-37.6	0.1	-33.5	-37.1	0.1	-33.4	17.8	-18.4	16.1	-19.7	2.0	80	1.057

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.9	19.8	14.1	17.4	12.8	15.3	11.7	14.5	18.6	13.1	16.7	11.9	14.9	5.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	8.9	16.3	11.1	8.3	14.8	10.1	7.7	13.8	40.5	18.5	37.1	16.8	34.0	14.9	18.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.5	11.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.5	-25.7	-24.2	-24.6	-15.8	-4.5	5.7	10.3	6.5	1.6	-5.6	-16.1	-22.2
	DBStd	14.27	7.95	8.97	7.63	6.51	5.67	4.47	3.65	3.64	3.68	3.98	6.43	7.43
	HDD10.0	7186	1106	958	1074	773	450	143	41	121	252	485	783	999
	HDD18.3	10151	1364	1192	1332	1023	708	379	248	367	502	743	1033	1257
	CDD10.0	77	0	0	0	0	0	13	51	13	0	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	6	0	0	0	0	0	1	6	0	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	5.4	5.6	5.6	5.1	5.7	5.5	5.8	5.3	5.2	5.2	5.8	5.0	5.1
Precipitation	PrecAvg	246	12	12	11	8	10	15	34	47	36	31	17	13
	PrecMax	478	42	44	96	29	34	53	65	93	132	71	60	56
	PrecMin	125	2	2	0	0	2	1	9	16	9	12	4	2
	PrecStd	83	10	10	17	6	8	10	15	22	25	15	11	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.2	-1.3	-0.6	0.8	7.8	21.6	23.6	20.2	14.9	4.0	0.2	-1.5
		MCWB	-1.6	-1.8	-1.4	-0.2	5.8	14.7	15.8	14.0	10.7	2.9	-0.1	-1.9
	2%	DB	-5.1	-4.3	-5.1	-2.3	5.7	17.9	20.7	16.9	11.6	2.2	-2.2	-4.1
		MCWB	-5.4	-4.7	-5.6	-2.9	3.8	12.2	14.6	12.6	8.7	1.6	-2.4	-4.6
	5%	DB	-11.0	-7.2	-9.2	-4.2	3.8	15.4	18.6	14.5	9.1	0.7	-4.9	-7.7
		MCWB	-11.3	-7.5	-9.5	-4.6	2.5	11.0	13.6	11.4	7.6	0.3	-5.3	-7.9
	10%	DB	-14.9	-10.3	-14.4	-6.6	2.3	13.0	16.6	12.5	7.0	-0.6	-7.0	-11.9
		MCWB	-15.1	-10.7	-14.8	-6.8	1.3	9.7	12.7	10.3	6.0	-1.0	-7.2	-12.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.1	-1.7	-1.0	-0.1	6.0	15.1	17.0	14.3	10.8	3.2	0.0	-2.0
		MCDB	-0.7	-1.3	-0.2	0.8	7.5	21.0	23.0	18.6	13.9	3.9	0.3	-1.7
	2%	WB	-5.1	-4.3	-5.3	-2.8	4.0	12.7	15.2	13.0	9.1	1.6	-2.2	-4.4
		MCDB	-4.7	-3.9	-4.9	-2.4	5.4	17.2	19.7	15.9	11.3	2.1	-2.1	-3.9
	5%	WB	-10.3	-7.0	-9.0	-4.5	2.5	11.2	14.1	11.7	7.7	0.4	-5.2	-7.7
		MCDB	-10.1	-6.6	-8.7	-4.2	3.7	14.8	17.6	14.1	9.1	0.8	-4.9	-7.5
	10%	WB	-14.1	-9.8	-13.9	-6.8	1.3	9.8	12.9	10.5	5.9	-0.9	-7.2	-12.0
		MCDB	-13.8	-9.5	-13.4	-6.6	2.2	13.0	16.3	12.4	7.0	-0.7	-7.0	-11.7
Mean Daily Temperature Range	5% DB	MDBR	6.9	7.4	6.9	7.2	5.0	7.6	7.9	6.1	4.5	4.7	6.3	6.8
		MCDBR	11.3	10.1	8.7	8.2	6.5	11.3	11.3	10.0	8.7	5.0	7.5	9.0
		MCWBR	11.2	10.0	8.5	8.0	4.9	6.8	6.2	6.2	6.0	4.5	7.4	8.9
	5% WB	MCDBR	11.4	10.0	8.6	8.2	6.4	10.8	10.4	9.1	8.6	4.8	7.6	9.0
		MCWBR	11.3	10.0	8.5	7.9	5.0	6.9	6.0	6.0	6.1	4.4	7.5	8.9
Clear-Sky Solar Irradiance	taub	N/A	0.208	0.244	0.305	0.324	0.327	0.338	0.325	0.286	0.208	N/A	N/A	
	taud	N/A	2.043	2.235	2.129	2.250	2.501	2.497	2.534	2.545	2.145	N/A	N/A	
	Ebn at Noon	0	514	778	816	851	868	841	811	747	508	0	0	
	Edh at Noon	0	37	67	105	107	87	85	72	52	31	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.32	1.82	4.10	5.47	5.66	4.79	2.95	1.45	0.41	0.00	0.00	
	RadStd	0.00	0.02	0.07	0.26	0.35	0.45	0.52	0.29	0.13	0.04	0.00	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		+1.46	+2.68	+1.38	N/A	N/A	N/A	-483	-485	N/A	N/A

Nomenclature: See separate page